22. Builder Design Pattern (Creational Design Pattern)

```
no usages
public class Student {
    1 usage
    int rollNumber;
    1 usage
    int age;
    1 usage
    String name;
    1 usage
    String fatherName;
    1 usage
    String motherName;
    1 usage
    List<String> subjects;
    1 usage
    String mobileNumber;

    no usages
    public Student(int rollNumber, int age, String name, String fatherName, String motherName, List<String> subjects, String mobileNumber){
        this.rollNumber = rollNumber;
        this.age = age;
        this.name = name;
        this.fatherName = fatherName;
        this.motherName = motherName;
        this.subjects = subjects;
        this.mobileNumber = mobileNumber;
    }
```

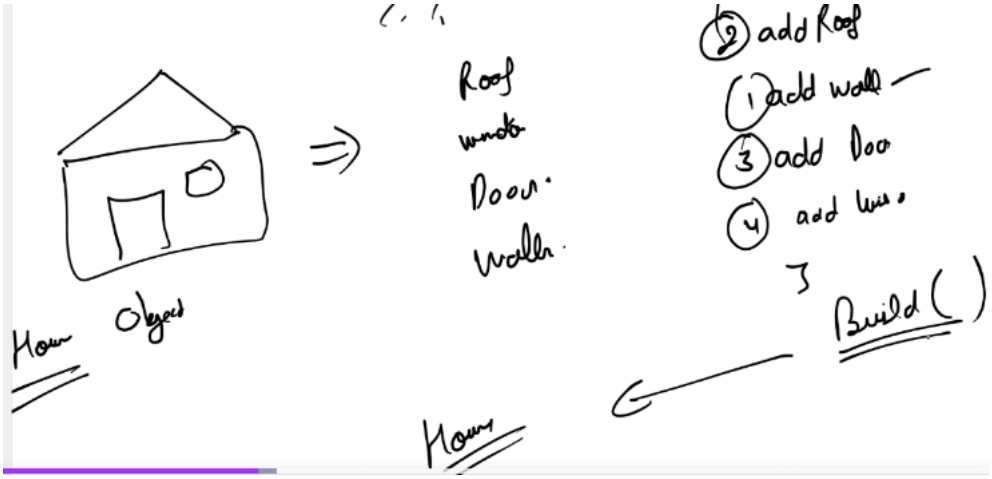
```
no usages
public Student(int rollNumber, int age){
    this.rollNumber = rollNumber;
    this.age = age;
}
no usages
public Student(int rollNumber, int age, String name){
    this.rollNumber = rollNumber;
    this.age = age;
    this.name = name;
}
no usages
public Student(int rollNumber, String fatherName, String motherName){
    this.rollNumber = rollNumber;
    this.fatherName = fatherName;
    this.motherName = motherName;
}
```

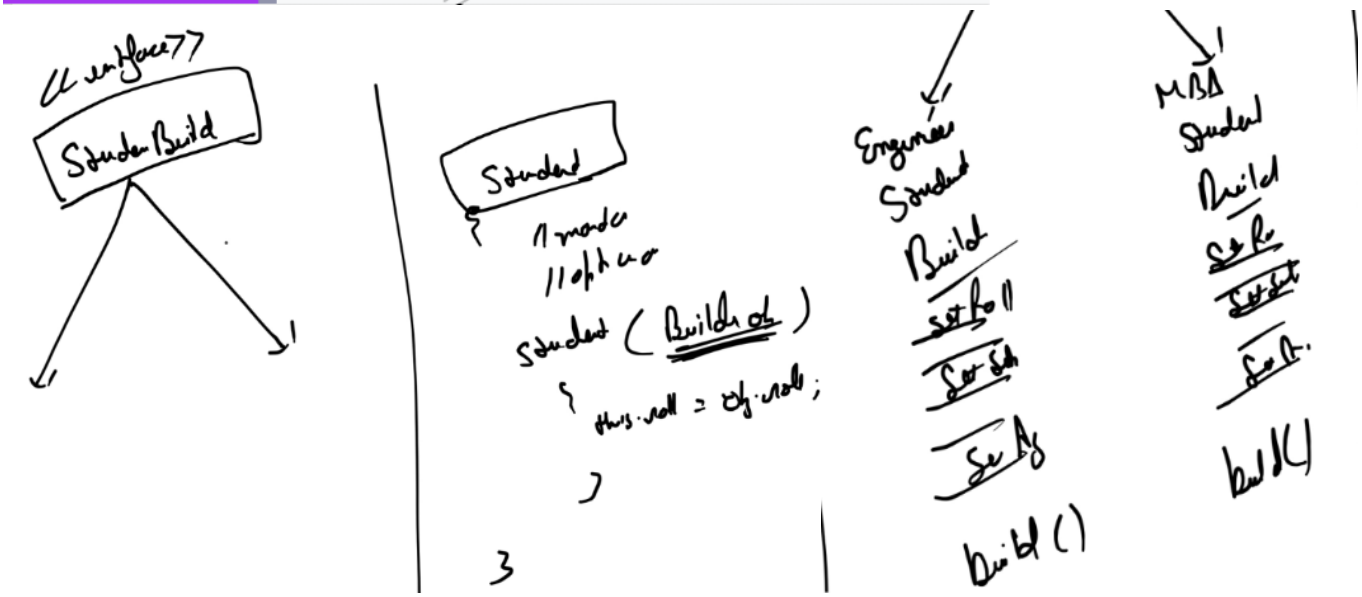
Number of Constructors can become huge

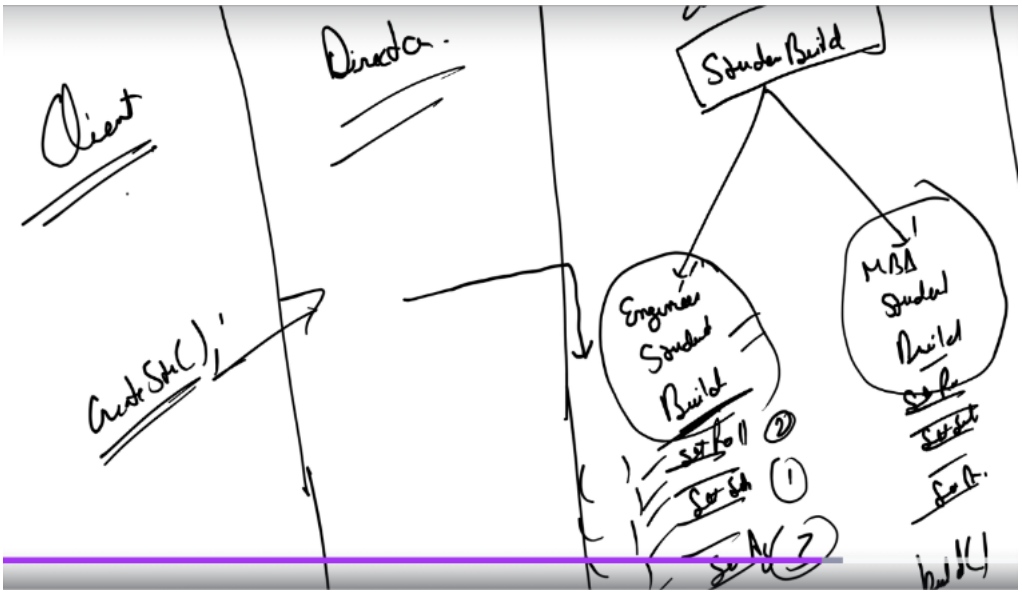
```
no usages
public Student(int rollNumber, String fatherName, String motherName){
    this.rollNumber = rollNumber;
    this.fatherName = fatherName;
    this.motherName = motherName;
}
```

Signature same - compilation error

https://gitlab.com/shrayansh8/lld-lowleveldesign/-/tree/main/src/main/java/com/conceptcoding/creationalpatterns/builder?ref_type=heads





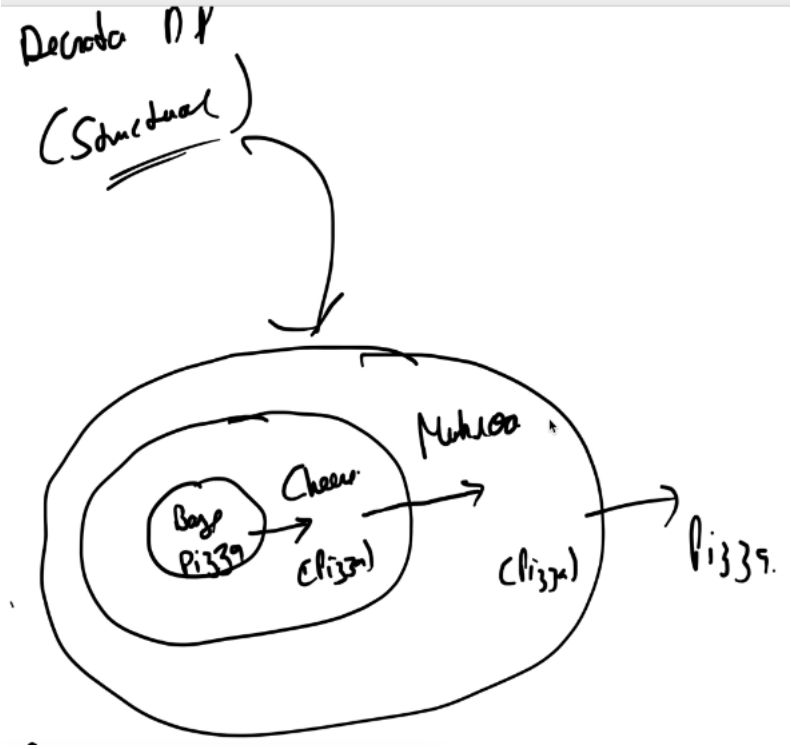


Builder

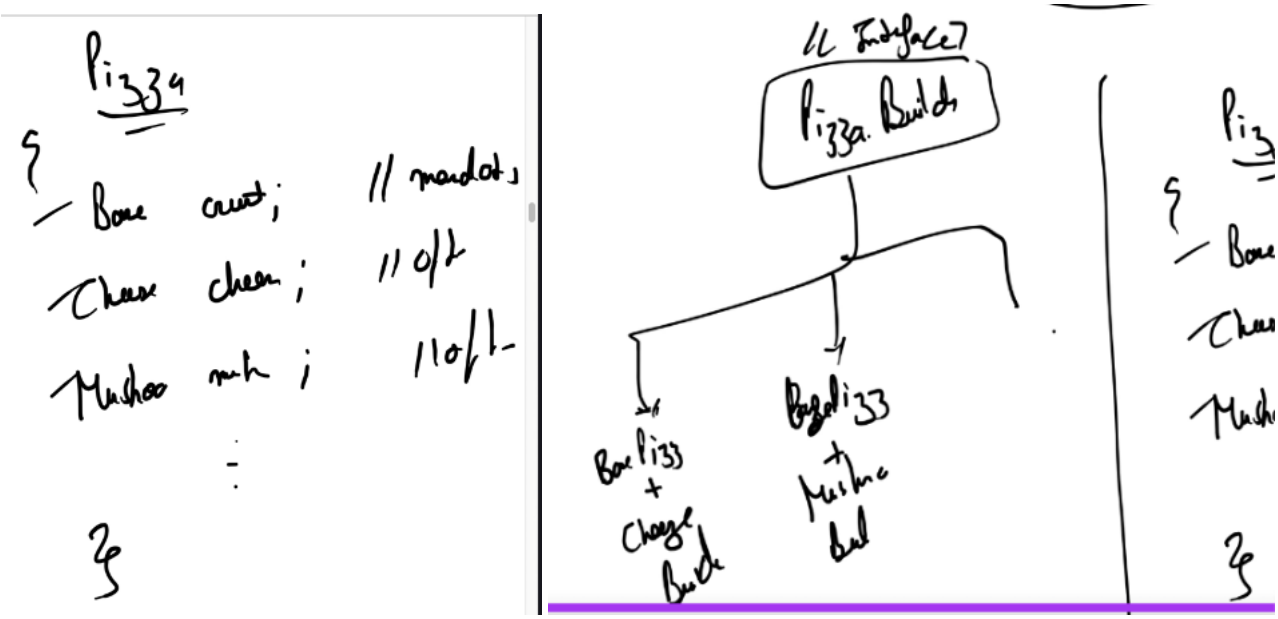
Creational

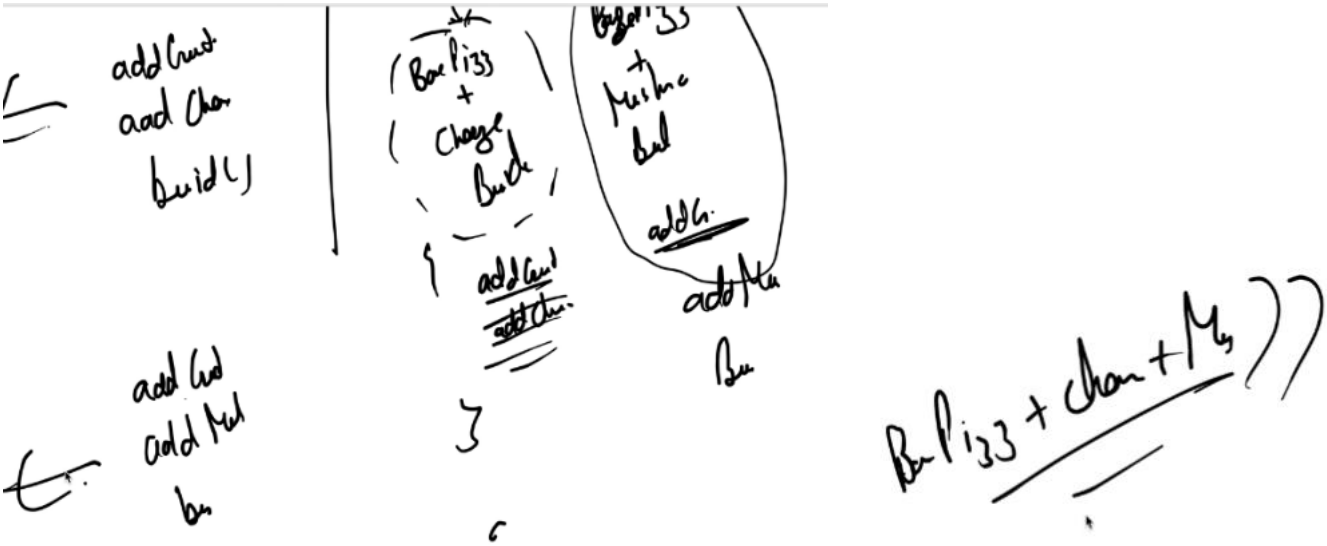
Decorator DP

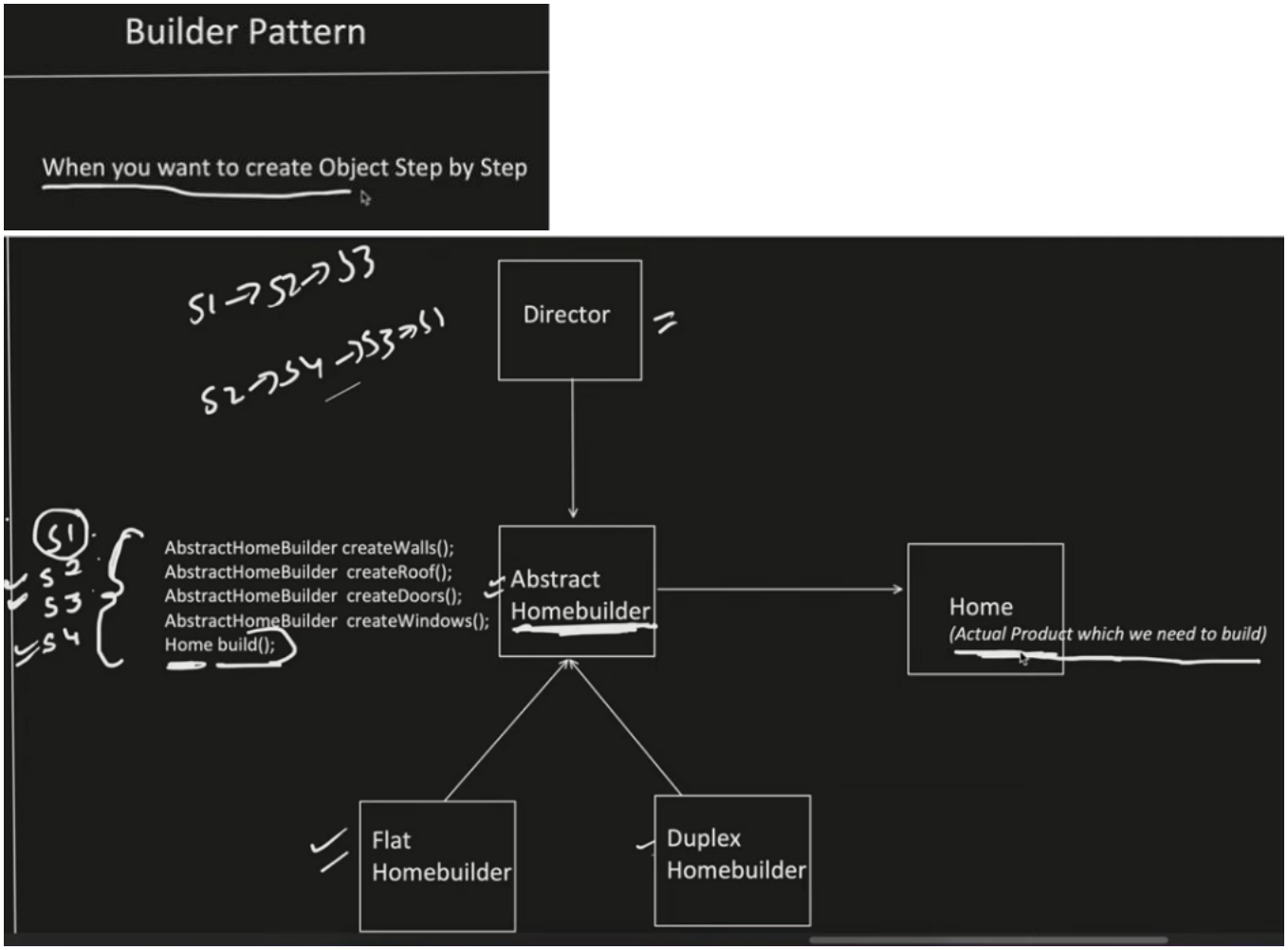
(Structural)



Pizza using Builder pattern :







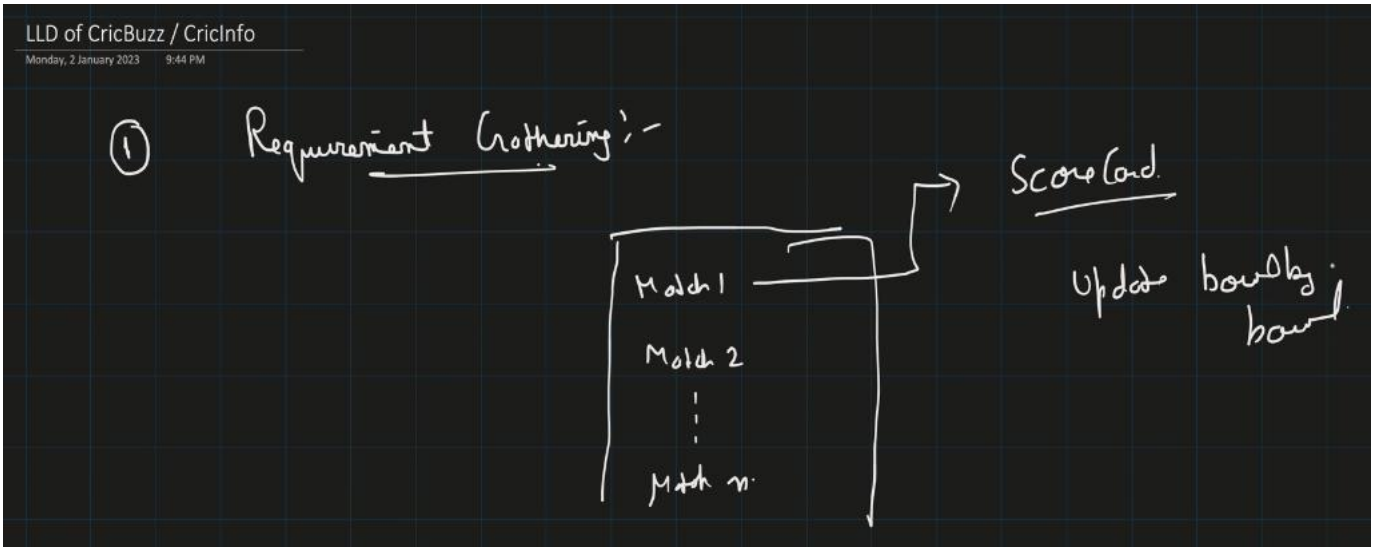
https://gitlab.com/shrayansh8/lld-lowleveldesign/-/tree/main/src/main/java/com/conceptcoding/creationalpatterns/builder?ref_type=heads

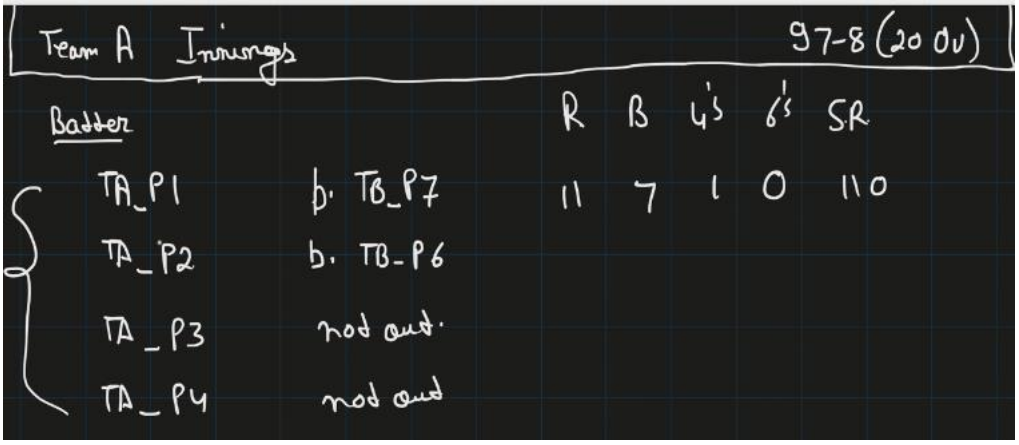
23. LLD of Cricbuzz

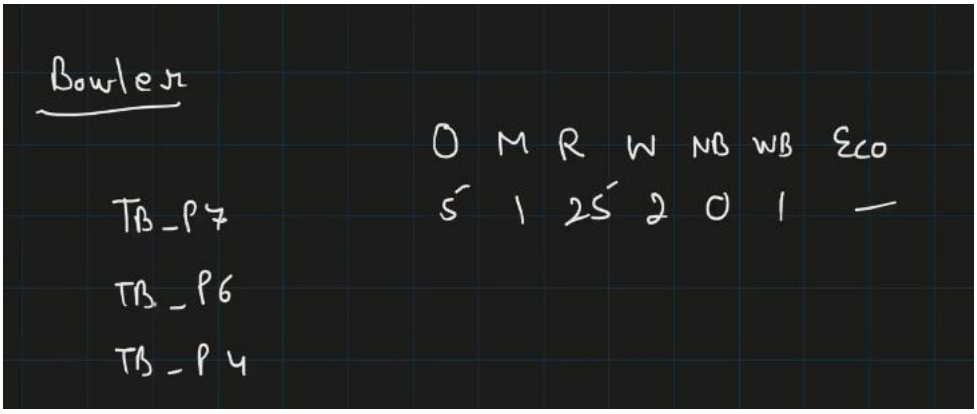
https://gitlab.com/shrayansh8/interviewcodingpractise/-/tree/main/src/main/java/com/conceptandcoding/LowLevelDesign/LLDCricbuzz?ref_type=heads

Low Level Design Roadmap

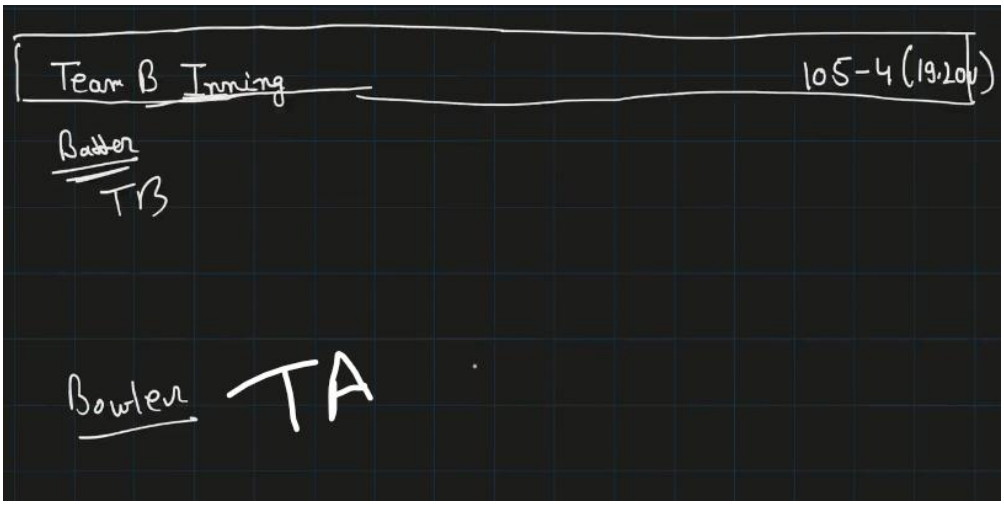
Low Level Design: Patterns to be covered	Low Level Design: Popular Interview Questions to be Covered
✓ Strategy Pattern	✓ S.O.L.I.D Principles
✓ Observer Pattern	✓ Design Notify-Me Button Functionality
✓ Decorator Pattern	✓ Design Pizza Billing System
✓ Factory Pattern	✓ Design Parking Lot
✓ Abstract Factory Pattern	✓ Design Snake n Ladder game
✓ Chain of Responsibility Pattern	✓ Design Elevator System
✓ Proxy Pattern	✓ Design Car Rental System
✓ Null Object Pattern	✓ Design Logging System
✓ State Pattern	✓ Design Tic-Tac-Toe game
✓ Composite Pattern	✓ Design BookMyShow & Concurrency handling
✓ Adapter Pattern	✓ Design Vending Machine
✓ Singleton Pattern	✓ Design ATM
✓ Builder Pattern	✓ Design Chess game
Prototype Pattern	✓ Design File System
Bridge Pattern	✓ Design Splitwise
Façade Pattern	✓ Splitwise Simplify Algorithm / Optimal Accounting Balancing
Flyweight Pattern	Design CricBuzz / CricketInfo
Command Pattern	Design True Caller
Interpreter Pattern	Design Car Booking Service like Ola, Uber
Iterator Pattern	Design Online Hotel Booking System
Mediator Pattern	Design Library Management System

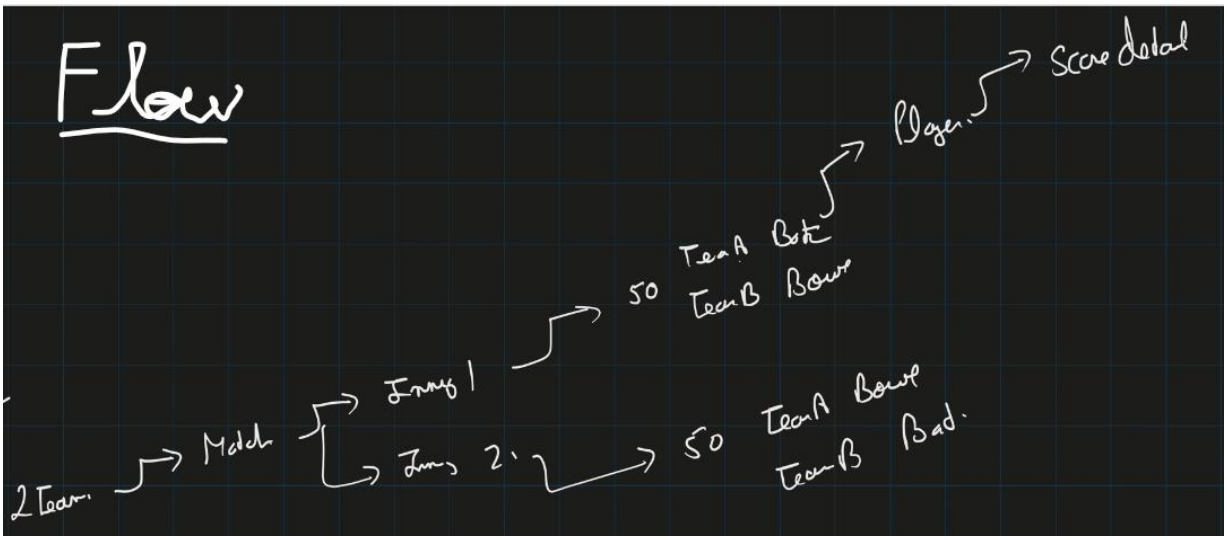




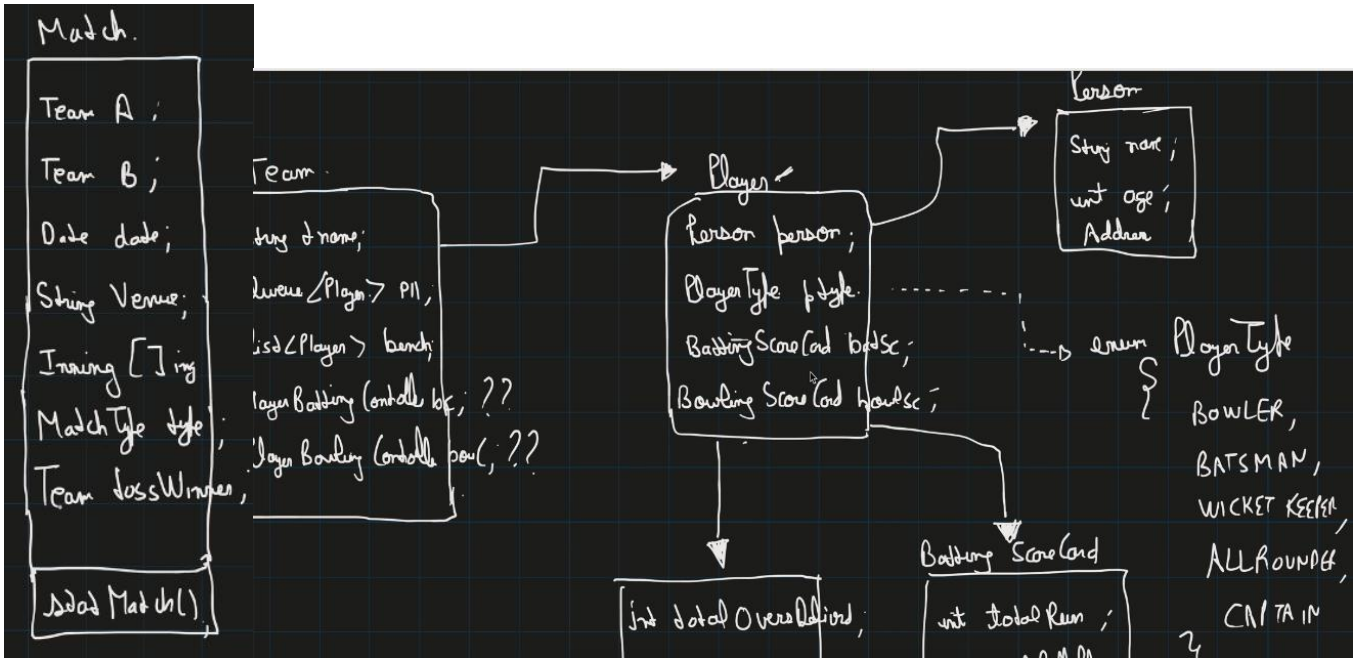


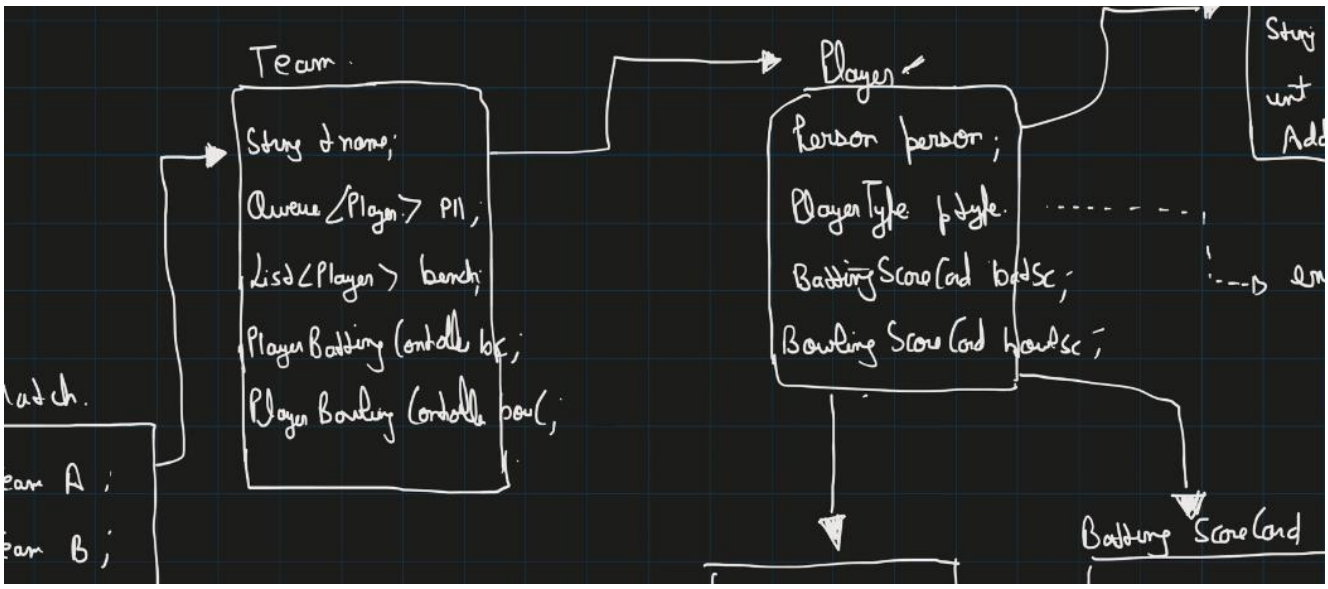
Team A Innings – Batting – team A, Bowling – team B

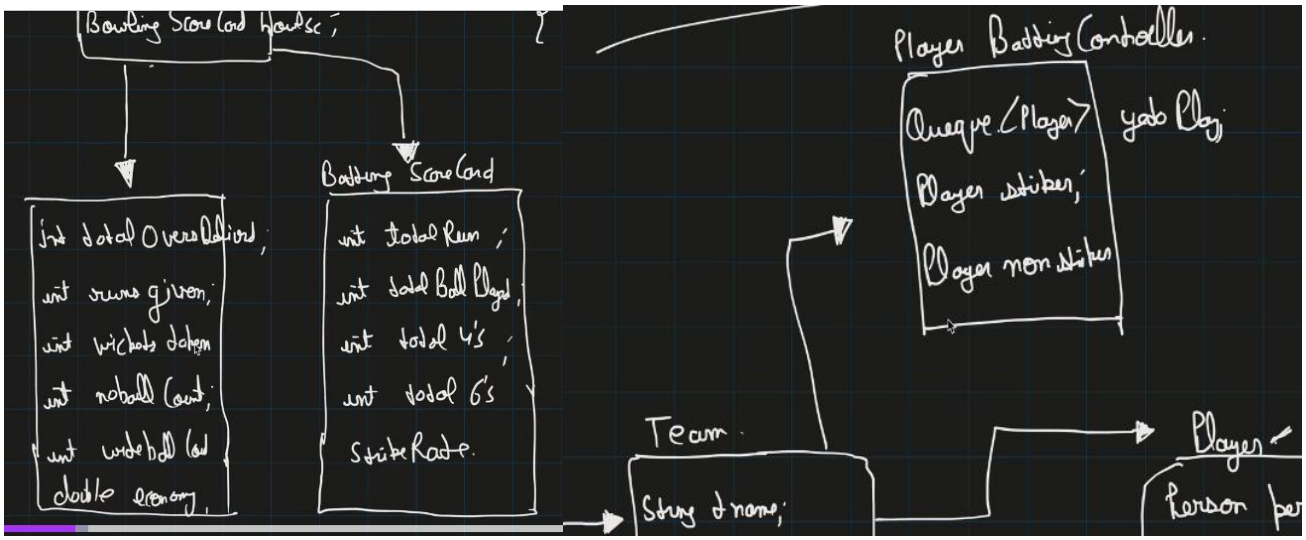


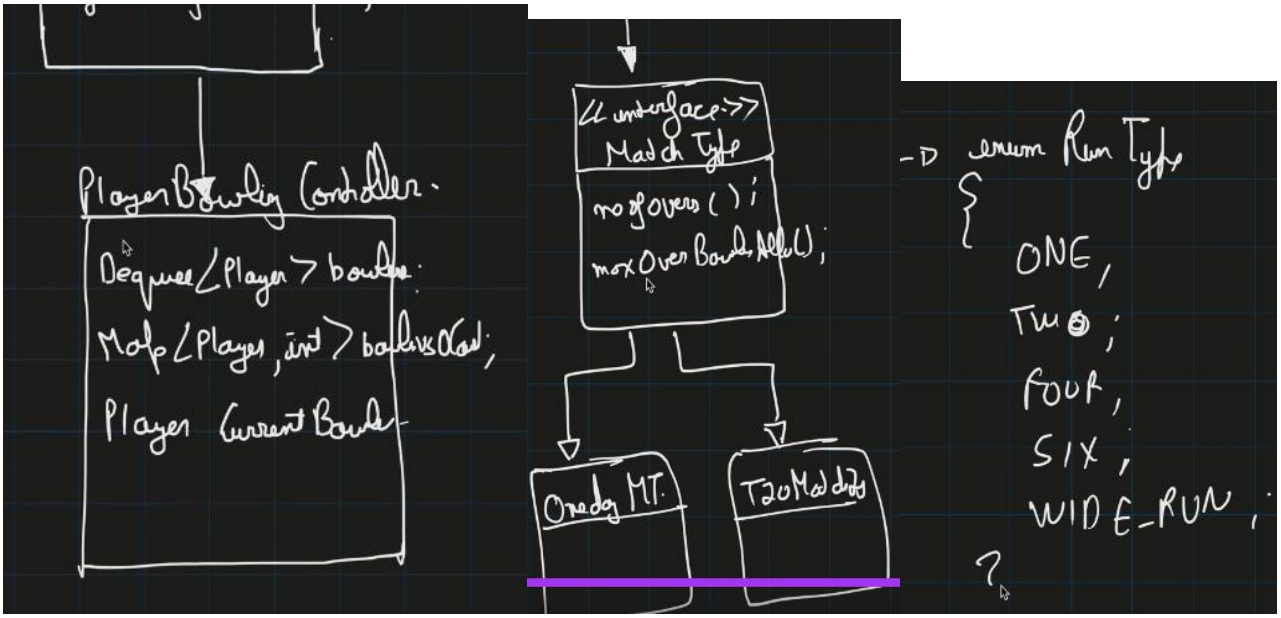


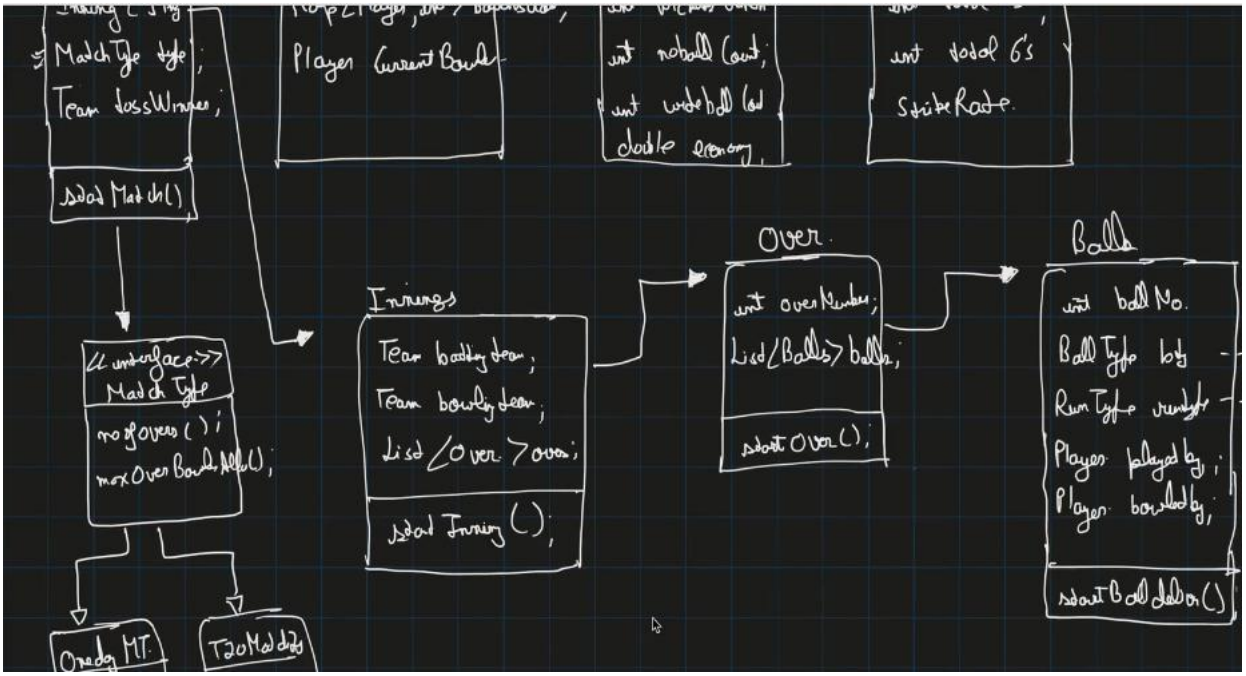
https://gitlab.com/shrayansh8/interviewcodingpractise/-/tree/main/src/main/java/com/conceptandcoding/LowLevelDesign/LLDCricbuzz?ref_type=heads

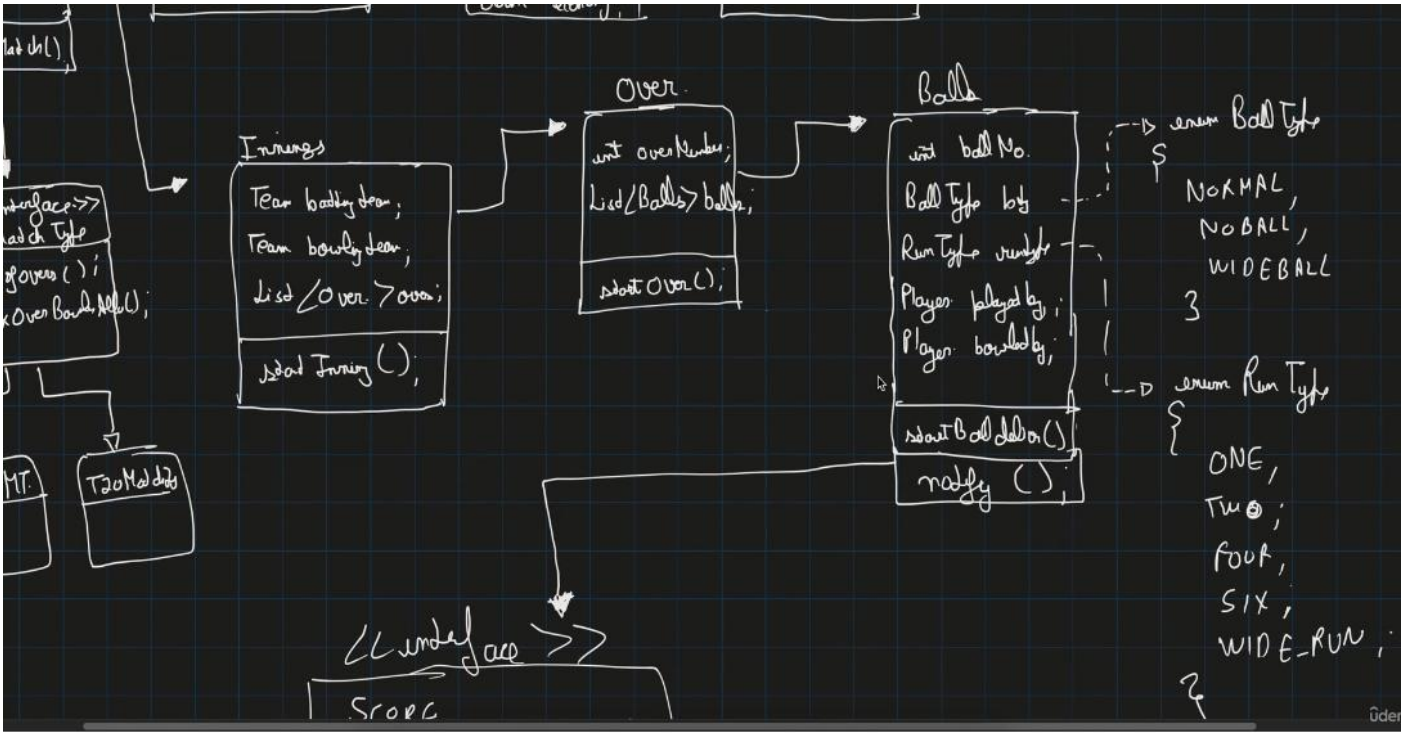

Match.
Team A ;
Team B ;
Date date;
String Venue;
Inning [] ing
MatchType type;
Team lossWinner;
startMatch()
Team.
String tname;
Queue<Player> PIL;
List<Player> bench;
PlayerBattingController bc; ??
PlayerBowlingController bowl; ??
Player
Person person;
PlayerType ptype.
BattingScoreCard bsc;
BowlingScoreCard bowlsc;
Person
String name;
int age;
Address
enum PlayerType
{
BOWLER,
BATSMAN,
WICKET KEEPER,
ALLROUNDER,
CAPTAIN
}
BattingScoreCard
int totalRun ;
int totalOversBowled;

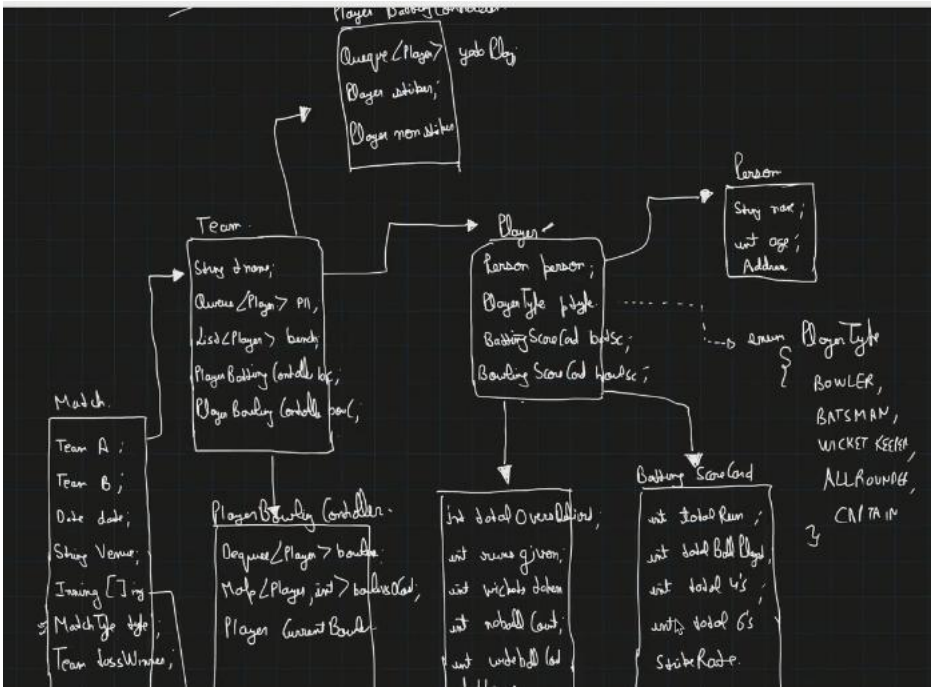

Team.
String tname;
Queue<Player> PIL;
List<Player> bench;
PlayerBattingController bc;
PlayerBowlingController bowl;
Player
Person person;
PlayerType ptype.
BattingScoreCard bsc;
BowlingScoreCard bowlsc;
BattingScoreCard

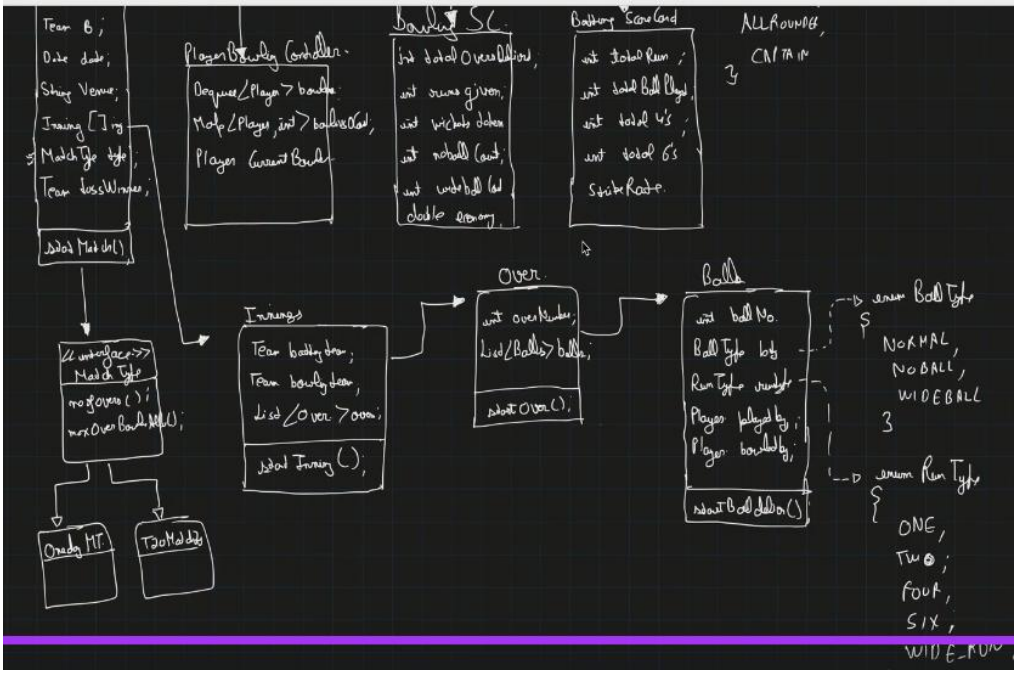

BowlingScoreCard bowlsc;
int totalOversBowled;
int runs given;
int wickets taken
int noballCount;
int wideballCount
double economy,
BattingScoreCard
int totalRun ;
int total Ball Played;
int total 4's ;
int total 6's
StrikeRate.
Team.
String tname;
PlayerBattingController.
Queque<Player> yet to Play;
Player striker;
Player non striker
Player
Person per

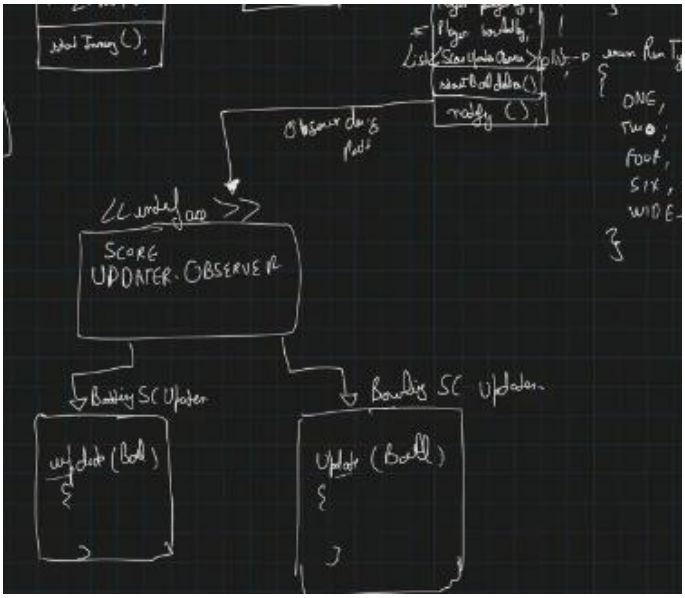

PlayerBowlingController.
Deque<Player> bowlers;
Map<Player, int> ballsOver;
Player currentBowler
<<interface>>
MatchType
noofovers();
maxOverBowlerAllow();
OnedayMT.
T20Matches
enum RunType
{
ONE,
TWO;
FOUR,
SIX,
WIDE_RUN;
?

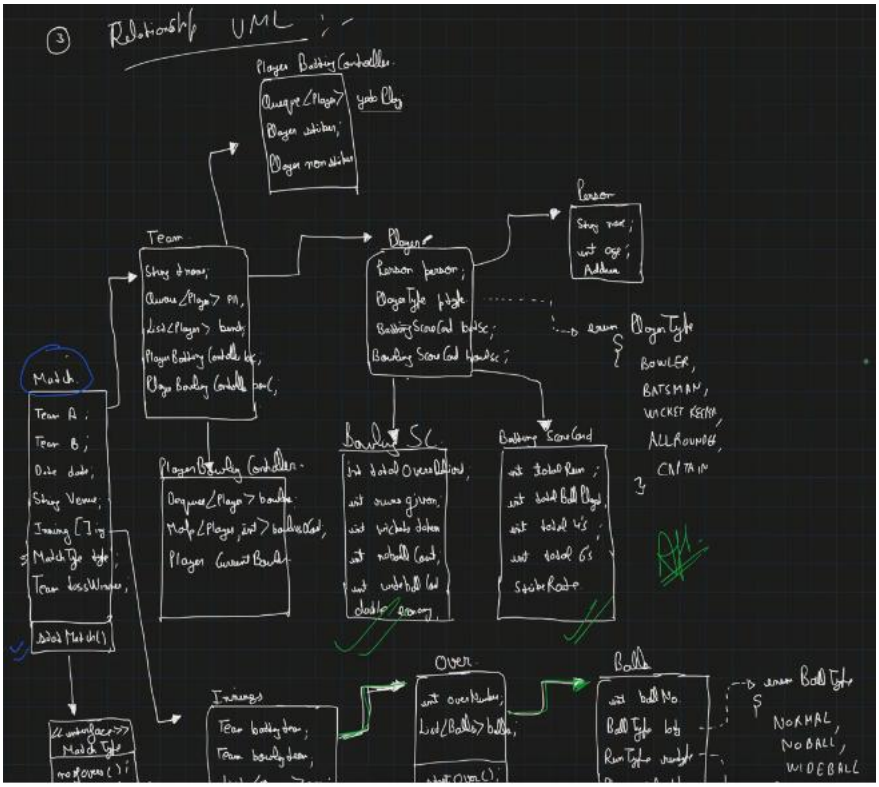

MatchType type;
TeamtossWinner;
startMatch()
Player currentBowler
int noballCount;
int wideballCount
double economy;
int totalG's
StrikeRate.
<<interface>>
MatchType
noofovers();
maxOverBowlerAllow();
OnedayMT.
T20Matches
Innings
Team battingTeam;
Team bowlingTeam;
List<Over> overs;
startInnings();
Over.
int overNumber;
List<Balls> balls;
startOver();
Balls
int ballNo.
BallType ty
RunType runtype
Player playedby;
Player bowledby;
startBallDelivery()

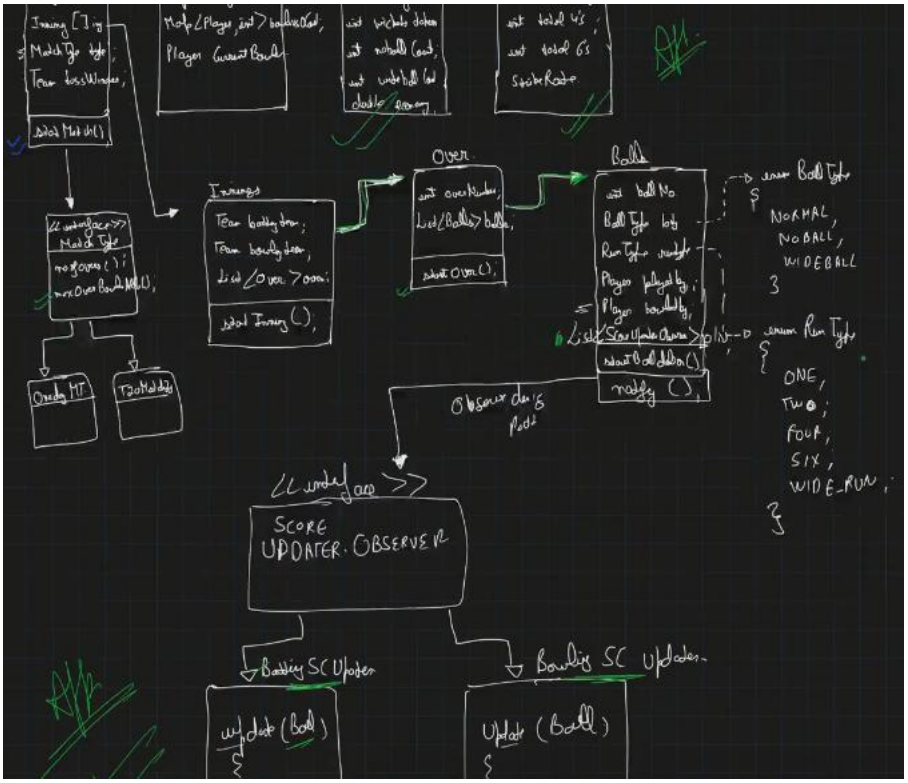

Over
int overNumber;
List<Balls> balls;
startOver();
Balls
int ballNo.
BallType bty
RunType runtype
Player playedby;
Player bowledby;
startBallDelivery()
notify ();
enum BallType
{
NORMAL,
NOBALL,
WIDEBALL
}
enum RunType
{
ONE,
TWO;
FOUR,
SIX,
WIDE_RUN;
}
Innings
Team battingTeam;
Team bowlingTeam;
List<Over> overs;
startInnings();
T20MatchDetails
<<interface>>
Score


Queue<Player> yetToBat;
Player striker;
Player nonStriker
Person
String name;
int age;
Address
Team
String teamName;
Queue<Player> PN;
List<Player> bench;
PlayerBattingController bc;
PlayerBowlingController bowl;
Player
Person person;
PlayerType ptype
BattingScoreCard bsc;
BowlingScoreCard bscc;
enum PlayerType
{
BOWLER,
BATSMAN,
WICKET KEEPER,
ALLROUNDER,
CAPTAIN
}
Match
Team A;
Team B;
Date date;
String Venue;
Inning [] inn
MatchType type;
Team tossWinner;
PlayerBowlingController
Dequeue<Player> bowler
Map<Player, int> ballsBowled;
Player CurrentBowler
int totalOversBowled;
int runs given;
int wickets taken
int noball Count;
int wideball Count
BattingScoreCard
int total Run;
int total Ball Played;
int total 4's;
int total 6's
StrikeRate


Team B;
Date date;
String Venue;
Inning [] inn
MatchType type;
Team lossWinner;
addMatch()
Player Bowling Controller
Deque<Player> bowler
Map<Player, int> bowlerOver
Player CurrentBowler
Bowling SC
int totalOversBowled
int runs given
int wickets taken
int noball count
int wideball ct
double economy
Batting Scorecard
int total Run
int total Ball Played
int total 4's
int total 6's
Strike Rate
ALLROUNDER
CAPTAIN
<<interface>>
MatchType
noOfOvers();
maxOverBowlerBall();
Creating MT
TossMatch
Innings
Team battingTeam;
Team bowlingTeam
List<Over> overs;
addInning();
Over
int overNumber
List<Balls> balls;
addOver();
Ball
int ballNo.
BallType type
RunType runtype
Player playedby;
Player bowledby;
addBallDetails()
enum BallType
NORMAL,
NOBALL,
WIDEBALL
enum RunType
ONE,
TWO,
FOUR,
SIX,
WIDE-RUN


addInning();
Player bowledby
List<StrikeUpdateBowler> pbl
addBallDetails()
notify()
Observer design Pattern
<<interface>>
SCORE UPDATER OBSERVER
Batting SC Updater
update(Ball)
Bowling SC Updater
Update(Ball)
ONE,
TWO,
FOUR,
SIX,
WIDE-






```
INNING 1 -- total Run: 133
---Batting ScoreCard : India---
PlayerName: India1 -- totalRuns: 0 -- totalBallsPlayed: 1 -- 4s: 0 -- 6s: 0 -- outby: SriLanka8
PlayerName: India2 -- totalRuns: 6 -- totalBallsPlayed: 2 -- 4s: 0 -- 6s: 1 -- outby: SriLanka8
PlayerName: India3 -- totalRuns: 2 -- totalBallsPlayed: 3 -- 4s: 0 -- 6s: 0 -- outby: SriLanka9
PlayerName: India4 -- totalRuns: 62 -- totalBallsPlayed: 17 -- 4s: 1 -- 6s: 8 -- outby: SriLanka9
PlayerName: India5 -- totalRuns: 1 -- totalBallsPlayed: 2 -- 4s: 0 -- 6s: 0 -- outby: SriLanka10
PlayerName: India6 -- totalRuns: 0 -- totalBallsPlayed: 1 -- 4s: 0 -- 6s: 0 -- outby: SriLanka10
PlayerName: India7 -- totalRuns: 27 -- totalBallsPlayed: 8 -- 4s: 1 -- 6s: 3 -- outby: SriLanka10
PlayerName: India8 -- totalRuns: 2 -- totalBallsPlayed: 2 -- 4s: 0 -- 6s: 0 -- outby: SriLanka9
PlayerName: India9 -- totalRuns: 12 -- totalBallsPlayed: 3 -- 4s: 1 -- 6s: 1 -- outby: notout
PlayerName: India10 -- totalRuns: 9 -- totalBallsPlayed: 4 -- 4s: 0 -- 6s: 1 -- outby: SriLanka11
PlayerName: India11 -- totalRuns: 12 -- totalBallsPlayed: 3 -- 4s: 0 -- 6s: 2 -- outby: SriLanka11

---Bowling ScoreCard : SriLanka---
PlayerName: SriLanka8 -- totalOversThrown: 2 -- totalRunsGiven: 46 -- WicketsTaken: 2
PlayerName: SriLanka9 -- totalOversThrown: 2 -- totalRunsGiven: 25 -- WicketsTaken: 3
PlayerName: SriLanka10 -- totalOversThrown: 2 -- totalRunsGiven: 35 -- WicketsTaken: 3
PlayerName: SriLanka11 -- totalOversThrown: 1 -- totalRunsGiven: 27 -- WicketsTaken: 2
```

```
LowLevelDesign.LLDCricBuzz.Match.Main
PlayerName: SriLanka2 -- totalRuns: 0 -- totalBallsPlayed: 1 -- 4s: 0 -- 6s: 0 -- outby: India8
PlayerName: SriLanka3 -- totalRuns: 21 -- totalBallsPlayed: 6 -- 4s: 0 -- 6s: 3 -- outby: India9
PlayerName: SriLanka4 -- totalRuns: 0 -- totalBallsPlayed: 1 -- 4s: 0 -- 6s: 0 -- outby: India8
PlayerName: SriLanka5 -- totalRuns: 14 -- totalBallsPlayed: 5 -- 4s: 0 -- 6s: 2 -- outby: India11
PlayerName: SriLanka6 -- totalRuns: 7 -- totalBallsPlayed: 3 -- 4s: 0 -- 6s: 1 -- outby: India10
PlayerName: SriLanka7 -- totalRuns: 33 -- totalBallsPlayed: 10 -- 4s: 1 -- 6s: 4 -- outby: India11
PlayerName: SriLanka8 -- totalRuns: 0 -- totalBallsPlayed: 1 -- 4s: 0 -- 6s: 0 -- outby: India11
PlayerName: SriLanka9 -- totalRuns: 6 -- totalBallsPlayed: 2 -- 4s: 0 -- 6s: 1 -- outby: India11
PlayerName: SriLanka10 -- totalRuns: 36 -- totalBallsPlayed: 11 -- 4s: 4 -- 6s: 2 -- outby: notout
PlayerName: SriLanka11 -- totalRuns: 10 -- totalBallsPlayed: 3 -- 4s: 0 -- 6s: 1 -- outby: notout

---Bowling ScoreCard : India---
PlayerName: India8 -- totalOversThrown: 2 -- totalRunsGiven: 28 -- WicketsTaken: 3
PlayerName: India9 -- totalOversThrown: 2 -- totalRunsGiven: 48 -- WicketsTaken: 1
PlayerName: India10 -- totalOversThrown: 2 -- totalRunsGiven: 29 -- WicketsTaken: 1
PlayerName: India11 -- totalOversThrown: 1 -- totalRunsGiven: 28 -- WicketsTaken: 4

---WINNER---SriLanka
```

https://gitlab.com/shrayansh8/interviewcodingpractise/-/tree/main/src/main/java/com/conceptandcoding/LowLevelDesign/LLDCricbuzz?ref_type=heads